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3D printed cervical standalone implant

Abstract

In one embodiment, an intervertebral implant includes a body and a locking element. The body includes a leading surface and a trailing surface opposite the leading surface. The body also includes first and second bone fastener passageways through the implant body and a cavity in between the first and second passageways. The cavity includes a trailing wall that separates the cavity from the trailing surface. The locking element is disposed in the cavity such that part of the locking element is visible through an access opening in the trailing wall so that the locking element may be rotated from outside of the implant. In a first rotational position, a first part of the locking element is located within one of the first and second passageways and in a second rotational position, the first part of the locking element is inside the body covered by the trailing wall.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 17/019,912, filed Sep. 14, 2020, which claims the benefit of the filing date of U.S. Provisional Patent Application No. 62/900,937, filed Sep. 16, 2019, the disclosures of which are hereby incorporated by reference herein in their entirety.

BACKGROUND OF THE INVENTION

(1) New implant technologies and associated methodologies are sought on an ongoing basis to improve outcomes of spinal surgery. When an implant is designed for use in an intervertebral space, the implant may serve to treat degenerative disc disease, misalignment of vertebral bodies, radiculopathy or myelopathy due to compression of the spine, or trauma (including that associated with a tumor), among other conditions.

(2) Past improvements to spinal implant technology have included improvements for interbodies with supplemental fixation, such as plates and screws or pedicle screws and associated components and in other cases, improvements for standalone intervertebral cages. Standalone cage designs typically include two, three or four screws for fixation of the cage to adjacent vertebrae. In more recent developments, some designs have incorporated a locking element to prevent back out of screws once the screws are disposed in the cage and anchored to bone.

(3) However, existing designs often have screw openings that provide a trajectory for the screws that undesirably directs the screws into a vertebra at a shallow angle. Further, to the extent existing designs may have protruding surfaces surrounding the screw openings in the cage to aid in

obtaining a desired screw trajectory, such designs often require the assembly of two separate components to form a complete cage. Additionally, existing designs with locking mechanisms may suffer from the locking mechanism becoming disengaged from a surface of the cage over time due to its open exposure on one side of the cage.

(4) Thus, a need exists for improved intervertebral implants and improved methods of manufacturing and implantation of intervertebral implants.

BRIEF SUMMARY OF THE INVENTION

(5) In one aspect, the present disclosure relates to an intervertebral implant. In one embodiment, the intervertebral implant includes an implant body and a locking element. The implant body includes a leading surface and a trailing surface opposite the leading surface. The implant body also includes first and second passageways through the implant body, each passageway sized for the disposal of a bone fastener therein. Further, the implant body also includes a cavity in between the first and second passageways, the cavity defined by a plurality of internal walls that include a trailing wall that separates the cavity from the trailing surface. The locking element includes a screw drive element and is disposed in the cavity such that the drive element is visible through an access opening in the trailing wall. The locking element is rotatable about its center through actuation of the screw drive element such that in a first rotational position, a first part of the locking element is located within one of the first and second passageways and in a second rotational position, the first part of the locking element is inside the implant body covered by the trailing wall.

(6) In some examples, the implant body may be monolithic. In some examples, the cavity may have a first perimeter measured in a plane parallel to the trailing surface and the access opening may have a second perimeter measured on the trailing surface. The first perimeter may entirely envelope the second perimeter when viewed facing the trailing surface. In some examples, the locking element may include a head that includes the first part of the locking element and a shaft that extends from the head. The shaft may engage with the body of the implant in the first rotational position and in the second rotational position. In some examples, the intervertebral implant may include a head that encompasses the first part of the first locking element. The head may have a width extending from a first end to a second end, the first end and the second end being inside the implant body and covered by the trailing wall in at least one rotational position of the locking element. In some examples, the shaft may include two separate parts extending longitudinally from the head of the locking element. In some examples, the locking element may include an internal cavity throughout its length along the central longitudinal axis.

(7) In some examples, the locking element may include a pin disposed in an opening through the locking element, the pin engaging directly with implant body so that the locking element is prevented from sliding out of the cavity. In some example, the first part of the locking element may be slidable into the implant through a second access opening in one of the passageways and in communication with the cavity. The first part of the locking element may be oriented so that the screw drive element is facing the trailing wall as the first part is inserted into the second access opening. In some embodiments, the first part of the locking element may be slidable into the implant through a second access opening on an inferior surface or a superior surface of the implant body. The second access opening may be in communication with the cavity. In some examples, the first passageway may be defined in part by a protrusion on an inferior surface of the implant body. In some examples, the second passageway may be defined in part by a protrusion on a superior surface of the implant body opposite the inferior surface.

(8) In some embodiments, the above described intervertebral implant may be formed through an additive manufacturing process. In particular, a method of forming the intervertebral implant may involve: forming the implant body of the intervertebral implant utilizing an additive layer manufacturing process; and inserting the locking element through a slot in the implant body, the slot being in communication with the cavity so that following insertion, the locking element is disposed in the cavity. In some examples, the method may include inserting a pin through the

locking element and into the cavity, the pin engaging with the implant body so that the locking element is secured to the implant body. In some examples, the additive layer manufacturing process may be performed based on the execution of software that retrieves geometric and material parameters of the implant body stored in a database. In some embodiments, forming the implant may involve a continuous, single step additive manufacturing process.

(9) In some embodiments, the above described intervertebral implant may be formed through an additive manufacturing process. In particular, a method of forming the intervertebral implant may involve: forming the implant body and the locking element of the intervertebral implant utilizing an additive layer manufacturing process. In some examples, forming the implant and the locking element may involve a continuous, single step additive manufacturing process. In some examples, the additive layer manufacturing process may be performed based on the execution of software that retrieves geometric and material parameters of the implant body and the locking element that are stored in a database.

(10) In another embodiment, the intervertebral implant includes a monolithic implant body and a locking element. The monolithic implant body includes a leading surface and a trailing surface opposite the leading surface along with a superior surface and an inferior surface opposite the superior surface. The implant includes a protruding part on the inferior surface or the superior surface. The protruding part defines a portion of a fastener opening that is sized for receipt of a bone fastener. The fastener opening extends from the trailing surface to the inferior surface or the superior surface. The implant also includes a central opening extending into the implant body from the trailing surface. Turning to the locking element, the locking element is disposable in the central opening of the monolithic implant body. The locking element includes a first engagement feature for engagement with a complementary second engagement feature of the implant body within the central opening. When the locking element is engaged to the monolithic implant body, the locking element is rotatable into a first rotational position where a portion of the fastener opening is covered by a portion of the locking element and a second rotational position where the opening is unobstructed by the locking element.

(11) In some examples, the implant may include a second protruding part, the second protruding part being on the inferior surface and defining a portion of a second fastener opening. The first protruding part may be on the superior surface. In some embodiments, the protruding part may form an arch shape over its length. In some embodiments, the fastener opening may be aligned along a first axis at an angle between 25 and 45 degrees relative to a central plane parallel to and in between the superior surface and the inferior surface. In some examples, the first axis may be at an angle of approximately 35 degrees relative to the central plane. In some examples, the second axis may be at an angle of approximately 35 degrees relative to the central plane.

(12) In some examples, the locking element may be centered on a central plane parallel to and in between the superior surface and the inferior surface and a center of the fastener opening at the trailing surface may be offset relative to the central plane. In some examples, the first fastener opening has a first center at the trailing surface and the second fastener opening has a second center at the trailing surface. The first center may be on a first side of a central plane parallel to and in between the superior surface and the inferior surface and the second center may be on a second side of the central plane. In some examples, the locking element may include a central cavity therein. The central cavity may extend longitudinally through a head and a shaft extending from the head and be sized for the receipt of an insertion instrument engagement feature. In some examples, the body may include a first plurality of channels extending inward from the leading surface and a second plurality of channels extending from the superior surface to the inferior surface. In some examples, the first plurality of channels may have a first pattern and the second plurality of channels may have a second pattern, the second pattern being different from the first pattern.

(13) In some embodiments, the intervertebral implant is part of a system that includes an insertion instrument. The insertion instrument may include a pair of prongs adapted to fit within the central

cavity of the locking element. The insertion instrument may also include a shaft advanceable into a space in between the pair of prongs to cause the pair of prongs to spread apart from one another and apply force against the locking element, thereby engaging the insertion instrument with the implant body.

(14) In another embodiment, the intervertebral implant includes an implant body and a locking element with flexible characteristics. The implant body includes a leading surface and a trailing surface opposite the leading surface. Within the implant body are first and second passageways, each passageway sized for the disposal of a bone fastener therein. The implant body also includes a recessed surface that is recessed relative to the trailing surface and has a length that extends from an edge of the first passageway to an edge of the second passageway. The recessed surface includes first and second dovetail grooves on sides of the recessed surface. Each dovetail groove extends from the edge of the first passageway to the edge of the second passageway. The locking element has a continuous perimeter with an opening therein, a width of the locking element being wider than a width of the recessed surface. Additionally, a length of the locking element is longer than the length of the recessed surface. The locking element is flexible such that it is insertable into the respective dovetail grooves of the implant body.

(15) In another aspect, the present disclosure relates to a spinal implant system. The system includes an intervertebral implant and an insertion instrument. The intervertebral implant of the system includes a body with a leading surface and a trailing surface opposite the leading surface. The implant also includes a first opening within the body sized for the disposal of a bone fastener therein, the first opening extending from the trailing surface to an inferior surface of the body or a superior surface of the body. Similarly, the implant includes a second opening within the body, the second opening extending into the body from the trailing surface. Additionally, the implant also includes a hollow locking element that is engaged to the body within the second opening. The hollow locking element is rotatable about its axis to block and unblock the first opening. Turning to the insertion instrument, the insertion instrument includes an outer shaft with a cannulated body and two longitudinally extending prongs extending from an end of the cannulated body, each prong having a reverse taper toward a respective free end. The hollow locking element also includes an inner shaft axially translatable within the cannulated shaft. Additionally, the insertion instrument includes an actuation mechanism adapted to control axial translation of the inner shaft. When the two longitudinally extending prongs are within a hollow part of the hollow locking element and the inner shaft is translated distally from a first position remote from the two longitudinally extending prongs to a second position in between the two longitudinally extending prongs, the two longitudinally extending prongs become further apart to engage the locking element.

(16) In some examples, the insertion instrument may also include first and second longitudinally extending arms positioned on opposite sides of the two longitudinally extending prongs, each longitudinally extending arm including an inward facing protrusion sized for engagement with a notch in the body of the intervertebral implant. In some examples, each of the first longitudinally extending arm and the second longitudinally extending arm may be biased so that respective distal ends of the arms are at a first distance from a central axis along the inner shaft and respective proximal ends of the arms are at a second distance from the central axis, the second distance greater than the first distance. In some examples, the body may include a first notch on a first side edge of the trailing surface and a second notch on a second side edge of the trailing surface, the notches adapted to receive the respective inward facing protrusions of the first and second longitudinally extending arms.

(17) In some examples, the insertion instrument may include a distal region adjacent to the end of the cannulated body, the distal region being wider than the cannulated body and including an inserter opening therethrough. The inserter opening may have a first central longitudinal axis at a first angle relative to the inferior surface of the body such that when the insertion instrument is engaged with the body, the first central longitudinal axis through the inserter opening is coincident

with a second central longitudinal axis through the first opening. In some examples, the locking element may be disposed at a center of the trailing surface. In some examples, the body may include a third opening within the body, the third opening sized for the disposal of a bone fastener therein and extending from the trailing surface to a superior surface of the body. The third opening may be positioned so that the second opening is in between the first opening and the third opening. In some examples, the system may include a drill guide slidably engageable with the insertion instrument. In some examples, the drill guide may include a first bore aligned at a first angle, and the first opening in the body may be aligned at the first angle. In some examples, the hollow part of the hollow locking element may be entirely within an actuatable drive element of the hollow locking element.

(18) In another aspect, the present disclosure relates to a method of manufacturing an intervertebral implant. In one embodiment, the method may involve steps including: forming a first portion of the intervertebral implant using additive layer manufacturing, the first portion including at least part of an internal cavity sized for disposal of a locking element therein; after forming the first portion, inserting the locking element into the internal cavity; and, after inserting the locking element, forming a second portion of the intervertebral implant using additive layer manufacturing, the second portion including a wall at least partially enclosing the locking element within the intervertebral implant.

(19) In some examples, the intervertebral implant may be formed through a subtractive form of manufacture. In some examples, the first portion and the second portion together may complete the formation of the intervertebral implant. In some examples, the method may include inserting a pin through the locking element and into the internal cavity of the intervertebral implant such that the locking element is prevented from disengaging from the intervertebral implant. In some examples, forming the first portion and the second portion may be based on details of the intervertebral implant design processed by an additive layer manufacturing machine.

(20) In another embodiment, a method of manufacturing an intervertebral implant may involve forming an intervertebral implant and a locking element together using additive layer manufacturing in a single continuous step. Subsequent to formation through the single continuous step, the locking element is disposed within the intervertebral implant within a cavity that is internal to an outer perimeter of the intervertebral implant.

(21) In yet another aspect, the present disclosure relates to a method of implanting an intervertebral implant into an intervertebral space. The method includes: inserting a locking element into a cavity of the intervertebral implant through a slot in an exterior surface of the intervertebral implant, a width of the cavity being greater than a width of the slot, the locking element being covered by an outer wall of the intervertebral implant once inserted; engaging an insertion instrument with the intervertebral implant; advancing the insertion instrument with the intervertebral implant into a prepared intervertebral space; positioning the implant within the prepared intervertebral space; inserting a bone fastener through a guide opening in the insertion instrument and subsequently through an opening in the intervertebral implant to anchor the bone fastener at an angle toward a corner of a vertebra adjacent to the prepared intervertebral space; removing the insertion instrument; and rotating the locking element to block the opening of the intervertebral implant to prevent back out of the bone fastener.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure will be better understood on reading the following detailed description of non-limiting embodiments thereof, and on examining the accompanying drawings, in which:

(2) FIGS. 1A-1C are various views of an implant with a locking element disposed therein

according to one embodiment of the disclosure.

(3) FIG. 1D is a sectional view of the implant and locking element of FIG. 1A.

(4) FIG. 1E is a sectional view of the implant of FIG. 1A.

(5) FIGS. 1F-1G are views of the locking element of FIG. 1A.

(6) FIGS. 2A-2C are various views of an implant with a locking element disposed therein according to another embodiment of the disclosure.

(7) FIG. 2D is a sectional view of the implant and locking element of FIG. 2B.

(8) FIGS. 2E-2F are views of the locking element of FIG. 2B.

(9) FIGS. 3A-3C are various views of an implant with a locking element disposed therein according to another embodiment of the disclosure.

(10) FIGS. 3D-3E are sectional views of the implant and locking element of FIG. 3A.

(11) FIGS. 3F-3G are views of the locking element of FIG. 3A.

(12) FIGS. 4A-4C are various views of an implant with a locking element disposed therein according to another embodiment of the disclosure.

(13) FIGS. 4D-4E are sectional views of the implant and locking element of FIG. 4A.

(14) FIGS. 5A-5C are various views of an implant with a locking element disposed therein according to another embodiment of the disclosure.

(15) FIG. 5D is a perspective view of the implant and locking element of FIG. 5A during assembly.

(16) FIG. 5E is a sectional view of the implant and locking element of FIG. 5A.

(17) FIGS. 6A-6C are various views of an implant with a locking element disposed therein according to another embodiment of the disclosure.

(18) FIG. 6D is a perspective view of the implant and locking element of FIG. 6A during assembly.

(19) FIG. 6E is a sectional view of the implant and locking element of FIG. 6A.

(20) FIGS. 7A-7C are various views of an implant with a locking element disposed therein according to another embodiment of the disclosure.

(21) FIG. 7D is a perspective view of the implant and locking element of FIG. 7A during assembly.

(22) FIG. 7E is a sectional view of the implant and locking element of FIG. 7A.

(23) FIGS. 8A-8C are various views of an implant with a locking element disposed therein according to another embodiment of the disclosure.

(24) FIG. 8D is a perspective view of the implant and locking element of FIG. 8A during assembly.

(25) FIGS. 9A-9C are various views of an implant with a locking element disposed therein according to another embodiment of the disclosure.

(26) FIG. 9D is a perspective view of the locking element of FIG. 9A.

(27) FIG. 10 is a side view of a locking element according to another embodiment of the disclosure.

(28) FIGS. 11, 12A-12B are various views of a locking element according to another embodiment of the disclosure.

(29) FIGS. 13, 14A, 14B are views of an implant inserter system and implant according to another embodiment of the disclosure.

(30) FIGS. 15 and 16 are views of an implant inserter system and implant according to another embodiment of the disclosure.

(31) FIGS. 17-18 are steps in a method of engaging an implant with the inserter of FIG. 13 according to another embodiment of the disclosure.

(32) FIGS. 19-20 are steps in a method of engaging an implant with an inserter according to another embodiment of the disclosure.

(33) FIGS. 21-24 are steps in a method of implanting an intervertebral implant into an intervertebral space according to another embodiment of the disclosure.

(34) FIGS. 25A-B are perspective and close up sectional views of a first step in a method of assembly of an intervertebral implant according to one embodiment of the disclosure.

(35) FIGS. 26A-B are perspective and close up sectional views of a second step in the method of

assembly of the intervertebral implant of FIGS. 25A-B.

DETAILED DESCRIPTION

(36) As used herein in reference to an implant, e.g., interbody cage, the term “superior” refers to a portion of the implant nearer the patient's head, while the term “inferior” refers to a portion of the implant nearer the user's feet, when the implant is implanted in an intended position and orientation. As with the terms “superior” and “inferior,” the term “anterior” refers to a portion of the implant nearer the front of the patient, the term “posterior” refers to a portion of the implant nearer the rear of the patient, the term “medial” refers to a portion of the implant nearer the mid-line of the patient, and the term “lateral” refers to a portion of the implant farther away from the mid-line of the patient. Additionally, the term “leading” refers to a portion of the implant that is inserted into the patient ahead of the remainder of the implant while conversely, the term “trailing” refers to a portion of the implant closest to an inserter instrument and is the last part of the implant inserted into the patient.

(37) In one aspect, the present disclosure relates to an implant including a locking element adapted for use in an intervertebral region within a spine. As shown and discussed, the implant is a standalone implant. Additionally, the implant is a monolithic structure and as such does not require assembly of separate parts to form an entirety of the structure. By way of example, the implant body may be monolithic with a separate locking element irremovably disposed therein. To form an implant in such a manner, including implants such as that shown in FIG. 2, employment of an additive manufacturing process is used because the combined implant and locking element cannot be manufactured using a subtractive manufacturing process. In some examples, additive manufacture may be layer-by-layer using an additive layer manufacturing (“ALM”), i.e., 3D printing, process. With formation of the implant using ALM, the need for assembly of multiple components is, in many instances, eliminated. Additionally, there is no need to build specialized inter-component engagement features into each of the separate components, as the implant is formed monolithically. In some examples, ALM processes are powder-bed based and involve one or more of selective laser sintering (SLS), selective laser melting (SLM), and electron beam melting (EBM), as disclosed in U.S. Pat. Nos. 7,537,664; 8,728,387; 9,180,010; and 9,456,901, the disclosures of which are hereby incorporated by reference in their entireties herein. Details of various step-by-step approaches to the ALM procedure that may be utilized to form the implant embodiments contemplated herein are discussed elsewhere in the present disclosure.

(38) In one embodiment, implant **100** is an intervertebral cage as shown in FIGS. **1A-1C** and includes a complementary locking element **150**. As shown in FIG. **1C**, the implant is generally shaped in the form a rectangular prism, though does include prominent protruding features adjacent to trailing surface **102**.

(39) Turning to the details of the external surfaces of implant **100**, the implant includes a superior surface **110**, an inferior surface **112**, side surfaces **106**, **108**, a trailing surface **102** and a leading surface **104**. Superior and inferior surfaces **110**, **112** are generally parallel to one another though taper slightly from the trailing surface toward the leading surface while trailing surface **102** and leading surface **104** are generally parallel to one another, as shown in FIG. **1C**. Side surfaces **106**, **108** taper toward one another from the leading surface toward the trailing surface, as shown in FIG. **1B**. The above described regions of the implant define a generally uniform profile, although adjacent to trailing surface **102**, the above described external surfaces include protrusions, as shown in FIGS. **1A** and **1C**. In particular, superior surface **110** includes a protrusion in the form of ridge **134** while inferior surface **112** includes a protrusion in the form of ridge **132**. These are described in greater detail below. It should be appreciated that in some alternative arrangements, the relationship between the various surfaces may vary from that shown. Additionally, each of superior surface **110** and inferior surface **112** includes a plurality of fins **114A**, **114B**, respectively, shown in FIG. **1C**. Each fin is oriented so that a length of the fin extends from one side surface to the other.

(40) Turning to the density of the implant structure, implant **100** includes lamellar features in the form of channels through a body of the implant, as best shown in FIGS. **1A-1C**. The channels as shown are diamond shaped and provide the implant with a customized density and/or variation in density over the implant volume that may mirror that found in certain bone structures, e.g., lamellar patterns. In alternative arrangements, the channels may be circular, another shape, or have more than one shape among the channels of the implant. Further examples and embodiments of implants with various channel configurations are described in U.S. Pat. Nos. 9,987,051, 10,028,841 and D824,518, the disclosures of which are hereby incorporated by reference herein in their entirety.

(41) In FIGS. **1A-1C**, implant **100** includes three sets of channels, each oriented orthogonally relative to the other two. A first grouping of channels are side channels **119** that extend approximately parallel with trailing and leading surfaces **102**, **104**. As shown in FIG. **1C**, side channels **119** are closely spaced and may be equidistant from one another measured in section approximately parallel to the side surfaces. Each channel is linear and includes a gap through central opening **113**, as shown in FIG. **1B**. In some examples, one or more channels may extend fully between the side surfaces of the implant. A second group of channels are vertical channels **118**, best shown in FIGS. **1A** and **1B**. Each vertical channel **118** is oriented in the superior-inferior axis when the implant is disposed in a spine. Channels may be approximately equally spaced from one another and are distributed around a body of the implant on all sides of opening **113**, as shown in FIG. **1B**. Some of the individual channels of the vertical channels **118** extend from a surface defining one of openings **122**, **124**, described in greater detail below, to one of superior surface **110** and inferior surface **112**, as shown in FIG. **1A**. Finally, a third group of channels **117** extend in a leading-trailing axis from leading surface **104** to central opening **113**, as shown in FIG. **1B**. Channels **117** may also be equally spaced from one another although as a group, as with the other channels, any pattern for the channels is contemplated. Additionally, channels may have varied spacing to fit around openings and other structural features of the implant body. The channels in the implant promote improved bone ingrowth and bone ongrowth. Further, inclusion of channels through the implant provides improved visualization through the implant both in person and via X-ray. For example, visualization is improved compared to solid titanium.

(42) Turning to the operative openings defining implant **100**, i.e., openings for bone fasteners and the locking element, the implant also includes a central opening **113** that extends through the implant from the superior surface to the inferior surface, as shown in FIG. **1B**. Central opening **113** is irregularly shaped, though an exact shape of the central opening may be different in alternative arrangements to accommodate particular surgical applications. The implant also includes a series of surfaces recessed from trailing surface **102** and leading to central longitudinal opening **111**. In turn, central longitudinal opening **111** is a passage extending an entirety of a distance between trailing surface **102** and central opening **113**. Central longitudinal opening **111** is centered coincident with a central longitudinal axis **182** of implant **100** and is sized for disposal of locking element **150** therein. As shown in FIG. **1B**, central longitudinal axis **182** is midway between side surfaces **106**, **108**.

(43) Turning to the details of the recessed surfaces, FIG. **1E** illustrates the trailing surface of the implant without the locking element. Immediately internal to trailing surface **102** is recessed surface **191**, the recessed surface having an arcuate perimeter in superior and inferior directions and following a path of openings **122**, **124** on sides separating the superior and inferior sides. The perimeter of recessed surface **191** is shaped to follow a footprint of the locking element through its range of rotational movement on the surface. This principle similarly applies to many of the other implants described herein. A depth of recessed surface **191** is sufficient so that head **152** of locking element **150** may be disposed therein. Within recessed surface **191** is opening **111**, with additional recessed areas extending outward from opening **111** immediately below recessed surface **191**. In particular, on one side of opening are indentations **192**, **193**, spaced apart from one another. On an opposite side is groove **196** with a surface that extends circumferentially around part of opening

111 from a first end **196A** to a second end **196B**. The above described surface features are sized and positioned to receive complementary features on locking element **150**. In particular, and as shown in FIG. **1D**, protrusion **163** of locking element is disposable in either indentation **192**, **193** and movable between the two by rotation of locking element **150** when the locking element is in the implant. Upper protrusion **168** slides within groove **196** between its ends **196A-B**. Details of locking element **150** are described below.

(44) Further to central longitudinal opening **111**, trailing surface **102** of the implant also includes two additional openings **122**, **124**, each located peripherally relative to central longitudinal opening **111**, as shown in FIGS. **1A** and **1E**. These openings are also interchangeably referred to as passageways throughout the disclosure. Each opening **122**, **124** extends into a body of the implant at an angle relative to a transverse plane **184** through the implant, as shown in FIGS. **1A** and **1C**. Transverse plane **184** is a central plane through the implant that is located in between the superior and the inferior surface. More specifically, a vector representing a central path of opening **122** from the trailing surface into the body of the implant is in an inferior direction away from transverse plane **184**, while a vector representing a central path of opening **124** from the trailing surface of the body into the body of the implant is in a superior direction away from transverse plane **184**. In this manner, openings **122**, **124** are angled in opposite directions relative to transverse plane **184**. Each opening **122**, **124** may be angled from approximately 25 to 45 degrees relative to transverse plane **184** when the locking element is disposed in the implant. Thus, an angle between a central axis through opening **122** and a central axis through opening **124** may be in a range from 50 degrees to 90 degrees. In some examples, each opening **122**, **124** is equal and opposite. For instance, each opening **122**, **124** may be angled at 35 degrees relative to transverse plane **184** for a total angle of 70 degrees between central axes of openings **122**, **124**. In other examples, one opening is at a 35 degree angle relative to transverse plane **184**. In at least some examples, each opening sized for fastener placement is parallel to the other when viewed in the transverse plane. Put another way, the openings remain at the same lateral distance from central longitudinal axis **182** over their length. The location of openings **122**, **124** on the implant and their associated angulation is advantageous in that it allows for fasteners disposed in the openings to engage a vertebra near one of its corners when the implant is disposed in an intervertebral space.

(45) Trailing surface **102** also includes notches **138A-B** on opposite sides of trailing surface **102** at side surfaces **106**, **108**, respectively. Each notch **138A-B** is approximately aligned with transverse plane **184**. Notches **138A-B** are sized for engagement by a tool, such as an insertion instrument, described in greater detail elsewhere in the present disclosure.

(46) To accommodate the advantageous location and angulation of openings **122**, **124** within implant **100**, implant **100** includes ridges **132**, **134** that protrude relative to other more planar surfaces of the implant. In light of the opposite direction of respective openings **122**, **124**, ridge **132** extends outward from inferior surface **112** while ridge **134** extends outward from superior surface **110**. Each ridge **132**, **134** defines part of the respective opening **122**, **124** entry on trailing surface **102**. As shown in FIG. **1A**, each ridge **132**, **134** is arcuate in shape with a concave surface facing inward toward a respective opening. The concave inside surface of each ridge **132**, **134** is angled to align with a pathway of a respective opening. For example, the inside surface of ridges **132**, **134** may define a partially cylindrical shape with a center becoming further from transverse plane **184** at locations of the implant further from trailing surface **102**. In another example, when opening **122** is angled at 35 degrees relative to transverse plane **184**, then a superior apex of the inner surface of ridge **132** may be at a 35 degree angle relative to the transverse plane. Each ridge **132**, **134** is shallow in depth and recedes to the primary superior and inferior surfaces **110**, **112**, close to trailing surface **102**, as shown in FIG. **1C**. The inclusion of ridges **132**, **134** on the implant provides geometry to allow for screw trajectories to place screws into corners of the adjacent vertebral bodies. By anchoring screws into corners of the adjacent vertebral bodies, bone purchase and stability is optimized. Additionally, ridges **132**, **134** may provide a stop surface to abut against

vertebral bodies when the implant is positioned in an intervertebral space. Also on trailing surface **102**, immediately superior and inferior to central longitudinal opening **111** are respective secondary protrusions **132A**, **134A** that define convex external surfaces. As shown in FIG. **1A**, the curvature of these protrusions is approximately parallel to that of a head of locking element **150**.

(47) Turning to the details of locking element **150**, the features of locking element **150** are best shown in FIGS. **1F** and **1G**. In particular, locking element **150** includes a head **152** and a shaft **166** extending from one side of the head to a free end **169**. Inside locking element **150** is a cavity **153** that extends through head **152** and into shaft **166**. The cavity is oriented and sized to accommodate receipt of a portion of a tool, such as insertion instrument **1300**, described in greater detail elsewhere in the present disclosure. At the entrance into cavity **153** at the head is a drive element **155** shaped to receive a driver (not shown) for actuation of the locking element. The driver may be a screw driver or other similar tool. Shaft **166** includes an open volume **165** that is carved out of part of a cylindrical surface of shaft **166**. Open volume **165** is partially enclosed by a flexible bar **161**. Flexible bar **161** extends from a first end at a base of open volume **165** and opposite a main portion of shaft **166**, to a second, free end closer to head **152**, as shown in FIG. **1F**. At the free end of flexible bar **161** is a protrusion **163** extending outward from flexible bar **161**. The inclusion of flexible bar **161** with protrusion **163** allows for locking element **150** to be rotated between at least two locked positions in the implant. For instance, to move from a first locked position to a second locked position, flexible bar **161** of locking element **150** is adapted to flex inward toward shaft **166** to bring protrusion **163** out of a first engagement feature, e.g., indentation **192**, within central longitudinal opening **111** of the implant, and locking element **150** is further adapted to be rotatable so that protrusion **163** may be rotated into a second engagement feature, e.g., indentation **193**, within central longitudinal opening **111**. In one position, head **152** of locking element **150** may partially block openings **122**, **124**. In another position, openings **122**, **124** may be unimpeded. In this and many other implant embodiments, rotational adjustment of the locking element is accompanied by visual, audible and tactile feedback to signal a change in a position of the locking element on or in the implant. The above described structural characteristics are described in greater detail in the method of use embodiments of the present disclosure.

(48) Locking element **150** also includes upper protrusion **168**, shown in FIG. **1F**, directly abutting head **152** and extending outward from shaft **166**. Upper protrusion **168** is adapted to be disposed in groove **196**, an arcuate-shaped recess in implant **100**, when locking element **150** is disposed in the implant. In particular, groove **196**, within recessed surface **191**, has an elongate dimension that limits a range of rotation of upper protrusion **168** by blocking its rotation beyond ends **196A**, **196B** of the arcuate shaped groove. In this manner, an overall rotation of locking element **150** is limited to a predetermined range.

(49) In one embodiment, an implant **200** and complementary locking element **250** are as shown in FIGS. **2A-2F**. For implant **200**, unless otherwise noted, like reference numerals refer to like elements of implant **100**, but within the **200** series of numbers. Implant **200** has generally the same outer shape as implant **100**, though central opening **213** is a different shape.

(50) In contrast with implant **100**, locking element **250** for implant **200** is disposed within an internal cavity **211** so that locking element **250** is not removable from the implant **200** after the implant is formed, a process described elsewhere in the present disclosure, e.g., ALM implant formation. The location of locking element **250** within implant **200** is best shown in FIG. **2B**. Beginning with the surface features of the implant surrounding internal cavity **211**, drive element **255** of the locking element is accessible from trailing surface **202** via a trailing access **225**. Trailing access **225** is an opening into cavity **211** that is aligned with a longitudinal axis **282**, although access **225** is smaller than head **252** of locking element **250**, thereby preventing locking element **250** from dislodging from implant **200** on the trailing side of the implant.

(51) Within cavity **211** is an outer part and an inner part. The outer part is defined by a volume between a trailing wall **203** on the trailing side of the implant and internal surface **291**, as shown in

FIGS. 2B and 2D. The volume of the outer part of cavity is sized for the rotatable disposal of head 252 of implant 200 therein. Superior and inferior sides of surface 291 are generally arcuate, though each opposing side includes a protrusion 288, 289, respectively, sized and positioned to prevent over-rotation of locking element 250. Between the opposing superior and inferior sides of surface 291 are openings in the form of side accesses 226, 227, or slots, that place cavity 211 in direct communication with openings 222, 224. In this manner, locking element 250 may be rotatable so that part of the locking element structure enters the passageways through openings 222, 224. The inner part of cavity 211 is recessed and internal to internal surface 291. In particular, internal to surface 291 are four indentations of which two, 292A and 292B, are shown in FIG. 2D. The indentations are positioned at approximately equal angles with respect to one another as measured from axis 282, and are each sized for the disposal of one of protrusions 263A, 263B therein. In this manner, in any fixed rotational position of locking element 250 within implant 200, protrusions 263A, 263B occupy two of the four indentations.

(52) Turning to locking element 250, shown in isolation in FIGS. 2E-2F, the locking element includes a head 252 and a shaft 266 extending from the head. Extending outward from the shaft on opposite sides are flexible bars 261A, 261B, separated from a main body of shaft 266 by open volumes 265A, 265B, respectively. Each flexible bar 261A, 261B is a cantilever and includes an exterior facing protrusion 263A, 263B at its respective free end. Protrusions 263A, 263B are sized for disposal in indentations 292A, 292B, and the other indentations internal to internal surface 291. Head 252 includes a first end portion 256 extending from a center of the locking element, i.e., at a center of drive element 255, and a second end portion 258 also extending from drive element 255, but in an opposite direction. Each end portion is offset from a superior-inferior axis centerline of locking element 250, though portion 256 is offset in a first direction while portion 258 is offset in a second, opposite direction. This is best shown in FIGS. 2D and 2F. One advantage derived from the housing of locking element 250 internally within cavity 211 is that locking element 250 is prevented from dislodging from the implant through the trailing face of the implant.

(53) In one embodiment, implant 300 and complementary locking element 350 are as shown in FIGS. 3A-3G. For implant 300, unless otherwise noted, like reference numerals refer to like elements of implant 200, but within the 300 series of numbers. In instances where the elements for implant 200 are not described, like reference numerals refer to like elements of implant 100, but within the 300 series of numbers.

(54) Turning initially to the structure sized for disposal of locking element 350 within implant 300, implant includes an inner cavity 311 with an outer part defined between a trailing wall 303 and surface 391 within implant 300, as shown in FIGS. 3C and 3E, and an inner part recessed relative to surface 391. The outer part is bounded by four separate and generally arcuate shaped walls, as shown in FIG. 3E, each wall separated from the other by accesses 326, 327, 328, 329. Side accesses 326, 327 provide for direct communication between inner cavity 311 and openings 322, 324 while superior and inferior accesses 328, 329 provide for direct communication with a superior side and an inferior side of the implant, respectively. Two of the walls defining outer part of inner cavity 311 oppose one another and include protrusions 388, 389, respectively, sized and positioned to limit rotational movement of locking element 350. Interior to internal surface 391 are indentations 392A-D, best shown in FIG. 3D. Each indentation extends internal relative to surface 391 as part of inner cavity 311.

(55) Locking element 350, shown in isolation in FIGS. 3F and 3G, is sized for disposal within implant 300 and is formed together with implant, e.g., via ALM, so that it is not detachable once the combined structure is formed. Locking element 350 includes head 352 and two flexible bars 361A, 361B extending therefrom. Each flexible bar is a mirror opposite of the other, as shown in FIGS. 3D and 3G. With reference to one flexible bar as representative, flexible bar 361A has an arcuate shape that is relatively uniform along its length as measured in a longitudinal axis of the locking element. On an internally facing surface of flexible bar 361A is a concave curved recess,

thereby providing a pathway through the implant from trailing surface **302** to central opening **313** even with locking element **350** disposed in the implant. On a central externally facing surface of flexible bar **361A** is a protrusion **363A**. Head **352** is sized for disposal in outer part of internal cavity **311** while protrusions **363A**, **363B** are each sized to snap-fit into any one of indentations **392A-B**, **393A-B**, a position of the locking element being adjustable through rotation of locking element within the internal cavity. One locked position of locking element **350** within implant **300** is illustrated in FIG. 3D.

(56) In one embodiment, implant **400** and complementary locking element **450** are as shown in FIGS. 4A-4E. For implant **400**, unless otherwise noted, like reference numerals refer to like elements of implant **200**, but within the **400** series of numbers. In instances where the elements for implant **200** are not described, like reference numerals refer to like elements of implant **100**, but within the **400** series of numbers.

(57) As with implant **200** and **300**, locking element **450** is entirely disposed within implant **400** in a manner such that locking element **450** is held within an internal cavity **411**. Turning to the structure of the internal cavity, internal to internal surface **491** is an inner part of internal cavity **411**, shown in section in FIG. 4D. The inner part is defined by a perimeter punctuated by protrusions **497B**, **497D**. Outer and inner parts of internal cavity **411** are sized for the disposal of head **452** and shaft **466** of locking element **450**, respectively.

(58) Locking element **450** includes head **452** and shaft **466** extending therefrom, as shown in FIG. 4B. Shaft **466** is a solid body with a partially cylindrical shape extending along a length from the head. The shaft includes troughs **464A-D**, each oriented in a lengthwise manner and spaced apart from the others. The geometry of the internal cavity **411** and locking element **450** are complementary such that locking element **450** is rotatable within the cavity into more than one locked setting whereby a trough, e.g., **464A**, may be moved from being unengaged with a protrusion to being engaged with protrusion **497D**. Such adjustments change an orientation of head **452** between a first orientation shown in FIG. 4E, where head **452** is blocking openings **422**, **424**, and a second orientation where head **452** is entirely over interior surface **491** and no longer blocks either opening **422**, **424**.

(59) In one embodiment, implant **500** and complementary locking element **550** are as shown in FIGS. 5A-5E. For implant **500**, unless otherwise noted, like reference numerals refer to like elements of implant **100**, but within the **500** series of numbers. In instances where the elements for implant **200** are not described, like reference numerals refer to like elements of implant **100**, but within the **500** series of numbers.

(60) In implant **500**, ridges **532**, **534** are reversed relative to the ridges of implant **100** so that ridge **532** protrudes on superior surface **510** while ridge **534** protrudes on inferior surface **512**. This configuration is commensurate with opening **522** being angled in a superior direction from the trailing end toward the leading end of the implant while opening **524** is angled in an inferior direction from the trailing end toward the leading end.

(61) Implant **500** includes an internal cavity **511** with an outer part sized to house locking element **550** and an inner part sized to receive pin **570**, as shown in FIGS. 5B and 5D. From trailing surface **502**, trailing access **525** through trailing wall **503** connects the trailing surface of the implant to internal cavity **511**. The outer part of internal cavity **511** extends between wall **503** and internal surface **591**, and is bounded between openings **522**, **524** by four side walls each separated by two of accesses **526**, **527**, **528**, **529**, as shown in FIG. 5E. Side accesses **526**, **527** provide space to allow ends of locking element **550** to rotate into openings **522**, **524**, while superior and inferior accesses **528**, **529** connect exterior surfaces of the implant with the internal cavity. Continuing to refer to the outer part of internal cavity, protrusions **588**, **589** provide a blocking surface that limits the extent to which locking element **550** is rotatable within the implant. Interior to internal surface **591** is inner part of internal cavity **511** including a second recessed volume **515A** and a third recessed volume **515B**, shown in FIGS. 5B and 5D. In variants, recessed volume **515B** may directly

communicate with internal cavity **511** without a second recessed volume. In this manner, pin **570** may be entirely enclosed within recessed volume **515B** below internal cavity **511**. As shown in FIG. **5B**, the internal cavity, as a whole, is sized to accommodate disposal of locking element **550** and pin **570** therein, where pin is disposed through locking element **550** and into third recessed volume **515B**. The pin and internal cavity are both shaped and surfaced so that pin **570** engages walls within third recessed volume **515B** through an interference fit.

(62) Locking element **550** is best shown in FIGS. **5D** and **5E** and includes end portions **556**, **558**, a central drive element **555**, and a central opening **553** through the drive element. Locking element **550** does not include a shaft. Rather, opening **553** is sized for the disposal of a pin **570** therein, as shown in FIG. **5B**. Additionally, side accesses **526**, **527**, or slots, in implant **500** are sized and positioned within respective openings **522**, **524** such that locking element **550** is insertable into cavity **511** by sliding it into one of side accesses **526**, **527**, as shown in FIG. **5D**. In this manner, locking element **550**, complemented by pin **570**, is advantageous in that it can be inserted into a formed implant even when the implant otherwise prevents disengagement of a locking element through trailing surface **502**. It should be appreciated that the assembly of locking element **550** with implant **500** takes place during manufacture.

(63) In one embodiment, implant **600** and complementary locking element **650** are as shown in FIGS. **6A-6E**. For implant **600**, unless otherwise noted, like reference numerals refer to like elements of implant **200**, but within the **600** series of numbers. In instances where the elements for implant **200** are not described, like reference numerals refer to like elements of implant **100**, but within the **600** series of numbers. Additionally, it should be appreciated that ridges **632**, **634** are reversed relative to the ridges of implant **100**, but that the ridge structures are otherwise the same as those described for implant **100** or contemplated alternatives.

(64) As with implant **500**, implant **600** includes an interior cavity **611** in communication with trailing surface **602** via trailing access **625**. Within interior cavity **611** is an outer part between wall **603** and internal surface **691** and an inner part within recessed volume **615**. A volume of upper part is defined by parallel walls that are diagonal to side surfaces **606**, **608** of implant **600**, as shown in FIG. **6E**, and includes blocking protrusions **688**, **689** to prevent over-rotation of locking element **650**. Outer part of interior cavity **611** is in communication with four access regions, i.e. slots, including side accesses **626**, **627**, superior access **628** and inferior access **629**. Each access **626**, **627**, **628**, **629** is fully enclosed on all sides and has dimensions sufficient so that locking element **650** may be slid from outside of the implant into cavity **611** via either opening **622**, **624** or via superior or inferior surfaces **610**, **612**. For example, in FIG. **6D**, locking element **650** is shown as ready for insertion through inferior access **629**. It should be appreciated that in alternative arrangements, the implant may include any sub combination of the four described access slots. For instance, three access openings in total: one on the superior surface and one in each fastener opening.

(65) In FIG. **6E**, the outer part of interior cavity **611** is sized to house locking element **650** with a pin **670** disposed therein, the pin further extending into recessed volume **615**. Locking element includes a drive element **655** and an opening **653** through the drive element, the opening at a center of the locking element and sized for receipt of pin **670**. As shown in FIG. **6D**, pin includes a lip, though pin may have alternative surface features chosen to suit the interconnectivity between the pin and the locking element. Pin **670** disposed in locking element **650** within implant **600** is shown in FIG. **6E**. Locking element **650** is held in position within implant **600** by pin **670**, which maintains an orientation of the locking element relative to the implant. Pin **670** is held fixed via an interference fit, i.e., press or friction fit, between pin **670** and walls of recessed volume **615**, although it is contemplated that other forms of interconnection between the pin and the implant may also be employed. As with implant **500**, locking element **650** is formed within or placed in implant **600** during manufacture.

(66) In one embodiment, implant **700** and complementary locking element **750** are as shown in

FIGS. 7A-7E. For implant **700**, unless otherwise noted, like reference numerals refer to like elements of implant **300**, but within the **700** series of numbers. In instances where the elements for implant **200** are not described, like reference numerals refer to like elements of implant **100**, but within the **700** series of numbers. Additionally, it should be appreciated that ridges **732**, **734** are reversed relative to the ridges of implant **100**, but that the ridge structures are otherwise the same as those described for implant **100** or contemplated alternatives.

(67) As with implant **500**, implant **700** includes an interior cavity **711** in communication with trailing surface **702** via trailing access **725**. Within interior cavity **711** is an outer part between trailing wall **703** and internal surface **791** and an inner part corresponding to recessed volume **715**. As shown in FIG. 7E, a volume of upper part is defined by three walls, one on a superior side of the upper part, another on a side facing opening **722**, and a third on a side facing opening **724**. The second and third walls are separated by inferior access **729**, while the first wall of the outer part is separated from the other walls by side accesses **726**, **727**. Each access **726**, **727**, **729** is a fully enclosed on all sides. The first wall and the third wall include blocking protrusions **788**, **789**, respectively, to prevent over-rotation of locking element **750**, when disposed in the implant. As noted above, outer part of interior cavity **711** is in communication with three access regions, including side accesses **726**, **727** and inferior access **729**. Access **729** has dimensions sufficient so that locking element **750** may be slid from outside of the implant into cavity **711**. In FIG. 7D, locking element **750** is shown as ready for insertion through inferior access **729**. It should be appreciated that in alternative arrangements, implant **700** may also include an access opening from the superior surface in addition to or as a substitute for inferior access **729**.

(68) As shown in FIGS. 7C and 7E, the outer part of interior cavity **711** is sized to house locking element **750** with a pin **770** disposed therein, the pin also extending further into an inner part of interior cavity **711**, the inner part being recessed volume **715** that extends directly into central opening **713** of implant. Locking element **750** includes an elongate slot **753** through its central axis with a widened opening **754** at its center, as shown in FIG. 7D. Pin **770** includes a head with a drive element **772** therein. In this manner, when pin **770** is engaged with locking element **750**, driving of the combined structure may be through the pin. Pin **770** also includes protrusions along its shaft, particularly immediately underneath the head, to hold pin rotationally fixed relative to locking element **750**. In this manner, rotation of drive element **772** when pin is disposed in locking element causes the combined pin and locking element structure to rotate together. When disposed in implant **700**, pin **770**, disposed through locking element **750**, is disposed within recessed volume **715** to form an interference fit with the implant body and locking element **750** is in turn held in place by pin. It should be appreciated that in alternative arrangements, other forms of interconnection between the pin and the implant may be employed. As with implant **500**, locking element **750** is formed within or placed in implant **700** during manufacture.

(69) In one embodiment, implant **800** and complementary locking element **850** are as shown in FIGS. 8A-8D. For implant **800**, unless otherwise noted, like reference numerals refer to like elements of implant **100**, but within the **800** series of numbers.

(70) Implant **800** includes a recessed surface **891** for receipt of a head **852** of locking element **850**, as shown in FIGS. 8A and 8B. Recessed surface **891** is shaped to accommodate rotation of locking element **850** when it is included therein, with protrusions **888**, **889** extending inward closer to central axis **882** to limit the rotation of locking element. Interior to recessed surface **891** is central longitudinal opening **811**, best shown in FIG. 8B, sized to accommodate disposal of a shaft of a locking element therein. Towards interior opening **813**, central longitudinal opening **811** includes a wider portion to match a shape of shaft **866** of locking element, which is wider at opposite ends, as shown in FIG. 8B. Because the head **852** at one end and a distal shaft portion at an opposite end are both wider than a central portion of shaft **866**, locking element is prevented from disengaging from implant **800** in both axial directions. A process of manufacturing implant **800** is similar to that utilized for implant **200**, and is described in greater detail elsewhere in the disclosure.

(71) Locking element **850** includes a head with offset end portions **856**, **858** extending to ends of the locking element and a drive element **855** at its center longitudinal axis, as shown in FIG. **8A**. Shaft **866** extends from the head and includes indents **864A-D** similar to those described for implant **400**. The indents on shaft **866** are sized and positioned so that locking element **850** may be rotatably fixed in different positions by moving an indentation into and out of one of protrusions **897A**, **897C** within central longitudinal opening **811**. For example, indentation **864A** may be moved from being on receiving protrusion **897A**, as shown in FIG. **8D**, to a position between protrusions **897A**, **897C** in a ninety degree rotation of locking element **850** about axis **882**.

(72) In one embodiment, implant **900** and complementary locking element **950** are as shown in FIGS. **9A-9D**. For implant **900**, unless otherwise noted, like reference numerals refer to like elements of implant **100**, but within the **900** series of numbers. Additionally, it should be appreciated that ridges **932**, **934** are reversed relative to the ridges of implant **100**, but the ridge structures are otherwise the same as those described for implant **100** or contemplated alternatives.

(73) Implant **900** includes a recessed surface **991** internal to trailing surface **902**, and recessed relative to recessed surface **991** is internal cavity **911**, shown in FIGS. **9B** and **9C**. Internal cavity **911** is generally of uniform size, though widens at its terminal end remote from trailing surface **902**. Recessed surface is sufficiently interior to trailing surface **902** so that locking element **950** head is disposable therein while internal cavity **911** is sized to receive a shaft in the form of flexible bars **961A**, **961B** extending from head **952** of locking element **950**. As shown in FIGS. **9B** and **9C**, locking element **950** is hollow with a central passage of sufficient size to accommodate disposal of pin **970** therein.

(74) As shown in FIG. **9D**, locking element **950** includes head **952** and a shaft **966** extending therefrom. Shaft **966** splits into two flexible bars **961A-B**, separated by a slit **965**. At free ends of each flexible bar are protrusions **963A**, **963B** extending outward from the shaft in opposite directions. Locking element **950** includes a drive element **955** and an opening or cannulation is internal to the drive element for disposal of a pin therein.

(75) In an assembled condition, with locking element **950** disposed in implant **900** via a snap fit, pin **970** is placed within locking element **950** and protrusions **963A-B** at a free end of the locking element are pushed outward against the walls of internal cavity **911**, thereby holding locking element **950** in place within the implant. The inclusion of pin **970** within locking element **950** prevents withdrawal of the locking element from the implant. Both locking element **950** and pin **970** provide an interference fit engagement.

(76) The implant and locking element may, together or independently, be varied in many ways. For example, in one embodiment, a locking element may engage an implant with a snap fit in a longitudinal direction instead of a radial direction, as with locking element **1050** shown in FIG. **10**. Locking element **1050** includes head **1052** and shaft **1066** extending therefrom. Within head **1052** is a flexible bar **1061** separated by the remainder of the head by slot **1065**. At a free end of flexible bar **1061** is a protrusion **1063** extending toward a free end of shaft **1066**. With locking element **1050**, screws in an intervertebral implant may be blocked by snapping locking element **1050** into exterior/outward facing notches or recesses in the implant. The locking element is adapted so that when rotated, flexible bar **1061** bends inward and narrows slot **1065** to disengage from a notch, then once rotated to another notch, snaps back into place.

(77) In another example, a locking element for an implant is in the form of a lever mechanism **1150** that is rotatable into a final position to hold a screw in place in the implant and prevent back out. For example, in FIGS. **11** and **12A-B**, lever mechanism is a two-part structure pivotable about a meeting point between the two parts. A first part **1152** and a second part **1154** of the lever mechanism are both U-shaped. The U-shaped bodies of lever mechanism **1150** are sized to be narrower than a head of a screw to be inserted into the implant. Second part **1154** is orthogonal to first part **1152**. At a free end of second part **1154** remote from the first part is an engagement feature **1155**. In some variations, a diameter of second part **1154** is less than a diameter of first part **1152**.

Lever mechanism **1150** is fixed to an implant **1100** at a juncture between the two parts, as shown in FIGS. **12A**, **12B** and is positioned so that it is rotatable until second part **1154** engages a wall of opening **1122** and engagement feature **1155** locks into place within a second engagement feature **1123**.

(78) In still further examples, two or more of the locking element structures and concepts described herein may be included on a trailing surface of an intervertebral implant to block, collectively, three or four or more bone fasteners. In other examples, a locking element may be used to block a single bone fastener.

(79) In another embodiment, an implant **1700** includes a locking element **1750** in the form of a wave spring, as shown in FIGS. **25A-26B**. For implant **1700**, unless otherwise noted, like reference numerals refer to like elements of implant **100**, but within the **1700** series of numbers. Implant **1700** is sized to receive locking element **1750** and includes a recessed surface **1791** on its trailing surface **1702**, as shown in FIG. **25B**, for example. Recessed surface **1791** extends between openings **1722**, **1724**, the openings being for receipt of fasteners within the implant. Along superior and inferior sides of recessed surface **1791** are grooves **1788**, **1789** shown in FIG. **25B**. Locking element **1750** is flexible and sized to snap into grooves **1788**, **1789**, as described in greater detail in the method. Locking element **1750** includes a widened central region with side portions **1751**, **1752**, and narrowed, loop ends **1756**, **1758**. Collectively, side portions and narrowed loop ends form a continuous, enclosed structure, as shown in FIG. **25A**. Locking element **1750** has a length such that, when overlaid or within trailing surface **1702**, each loop end **1756**, **1758** covers a respective opening **1722**, **1724** in the implant. Further, when the locking element is not compressed, a dimension between outside edges of side surfaces **1751**, **1752** is greater than a width into recessed surface **1791**. On respective side portions **1751**, **1752** are inward facing enclosed structures that encircle holes **1753**, **1754**. Holes **1753**, **1754** are sized for engagement with a holding instrument.

(80) In another aspect, the present disclosure relates to an insertion instrument. In one embodiment, insertion instrument **1300** is as shown in FIGS. **13**, **14A**, **14B** and **17-18**. With reference to FIG. **13**, insertion instrument **1300** includes a handle **1302** and an outer shaft **1311** extending from the handle. Additionally, within the handle and outer shaft is an axially translatable inner shaft **1313** shown in FIGS. **17-18**. At a distal end of insertion instrument **1300**, opposite handle **1302**, are engagement features for engagement to an implant, such as implant **100** shown engaged to the insertion instrument in FIGS. **13**, **14A**, **14B** and **17-18**. Outer shaft **1311** is generally tubular and extends to a distal end. From the distal end of outer shaft are first and second longitudinally extending prongs **1314A**, **1314B**, both of which extend to free ends, as shown in FIGS. **17** and **18**. Toward the free ends of first and second prongs **1314A**, **1314B**, each prong becomes wider in a direction facing the opposite prong. These widened tips are denoted by reference numerals **1315A**, **1315B**, respectively. Through this structure, each prong **1314A**, **1314B** at the distal end of the instrument enlarges upon advancement of inner shaft **1313** into the distal region of the instrument, thereby promoting engagement between the instrument and the subject of its engagement, such as an intervertebral implant. In an alternative configuration, the tubular outer shaft may extend to the distal end and include circumferentially spaced slots so that the entire tubular structure of the outer shaft expands upon receipt of the inner shaft as it advances distally. In further variations, the reverse-taper of the prongs may be substituted with protrusions having other shapes that provide an increased thickness at the distal end of the insertion instrument.

(81) With continued reference to a distal region **1317** of insertion instrument **1300**, distal region includes a widened body with angled fastener openings **1322**, **1324** passing therethrough, as best shown in FIG. **14A**. The fastener openings are positioned and angled to correspond to openings in the implant to be implanted, e.g., openings **122**, **124** for implant **100**. This arrangement is advantageous in that bone fasteners may be inserted into an implant while the insertion instrument is engaged to the implant, with space available to do so as shown in FIG. **14A**. Peripherally and

extending longitudinally on the outsides of fastener openings are alignment arms **1342**, **1344**, each including an inward facing prong (not shown) at a respective free end. While engaged to an implant, such as implant **100**, the prongs of arms **1342**, **1344** engage within notches **138A**, **138B** of the implant. The arms are advantageous in that while engaged to notches in the implant, the rotation of the implant relative to the insertion instrument is minimized.

(82) Axial translation of inner shaft **1313** is controllable through a position of lever arm **1334** relative to handle **1302**. In particular, rotation of lever arm **1334** about a pivot point at base **1333** causes inner shaft **1313** to move axially. Because the pivot point for lever arm is offset relative to a connection point between the lever arm and inner shaft **1313**, rotation of the lever arm causes inner shaft to be pushed or pulled with respect to the base of the lever arm upon its rotation.

(83) In one specific example of the actuation mechanism for the insertion tool, the actuation mechanism and the lever arm are connected to one another through an internal link. The internal link is connected to the inner shaft at one end and the lever arm at the other end, with a pin connection at both ends. The lever arm **34** is connected to the handle via a third pin separate from the first and second pins, and located at a location closer to an internal end of the lever arm than the pin for the internal link. Through the pin connection, the lever arm is pivotable about an axis through the third pin. In certain additional examples, a bottom surface of the handle opposite the lever arm includes a lock button that is secured to the handle via a pin that is threaded into a corresponding thread in a ball detent, disposed internally within the handle. The ball detent is disposed within the handle such that it lies immediately proximal to an internally disposed portion of lever arm. Proximal to the ball detent is a spring and then an end cap closing the enclosed channel of the handle at an end of insertion instrument. Through this assembly, the ball detent is axially adjustable from a biased position abutting the lever arm to a retracted position, with spring compressed, that is spaced apart from the lever arm. Through operative connection of the ball detent with the lock button, lock button is actuatable to retract the ball detent. This specific configuration provides a mechanism to control whether the lever arm may be rotated. When the lock button is retracted, the lever arm may be rotated. When the lock button is not, then the lever arm is locked in place. Thus, the operative connection between the lever arm and the inner shaft is such that rotation of lever arm **1334** about the base causes inner shaft **1313** to move axially either distally or proximally.

(84) Toward a distal end of outer shaft **1311** is a button **1320** secured thereto via internal springs (not shown) internal within the outer shaft. Without load applied to button **1320**, button **1320** is biased in a raised position relative to a surface of outer shaft **1311**, as shown in FIG. **13**, for example. However, button **1320** may be depressed with the application of forces thereon, thereby compressing the internal springs. One function of the button is to prevent axial movement of drill guide **1350** in a proximal direction when attached to insertion instrument **1300**, as shown in FIG. **14B**. In some examples, outer shaft **1311**, towards its distal end, also includes a pair of engagement features on opposing lateral sides that are in the form of longitudinally extending slots. These slots are sized for the disposal of engagement features of a drill guide **1350** therein.

(85) In another embodiment, insertion instrument **1400** is shown in FIGS. **19-20**. Insertion instrument **1400** includes a cannulated outer shaft **1409** with distally extending end arms **1442**, **1444** that extend longitudinally from an end region of the outer shaft. Each end arm **1442**, **1444** is pivotably connected to the outer shaft so that inward facing protrusions **1443**, **1445** at the ends of the respective arms **1442**, **1444** may be actuated through pivoting of arms **1442**, **1444**. A sleeve slidable over arms **1442**, **1444** may be included to function as an actuation mechanism, the sleeve sliding along a length of the arms to control their orientation relative to a central axis of the instrument. In an alternative arrangement, arms maybe flexibly connected to the outer shaft and bend in an outward direction upon application of force from an inside of the arms. In such an arrangement, the arms may be biased to be parallel with the central axis of the instrument or biased slightly inward. Arms **1442**, **1444** may extend from outer shaft **1409** at respective hinge points, e.g.,

1442A, **1444A**, respectively. It should be appreciated that the actuation function of arms **1442**, **1444** may be derived from a variety of arrangements. For example, from active control, the material properties of the arms themselves or from a living hinge, among others. Internal to outer shaft is cannulated intermediate shaft **1411**, and internal to intermediate shaft **1411** is inner shaft **1413** with a head **1415** extending radially outward relative to a central longitudinal axis of inner shaft **1413**. In some examples, head is closer in size to inner shaft **1413**. In those and other examples, head may be threaded. In other alternatives, any number of inner or outer sleeves and or shafts may be included with the instrument.

(86) As shown in FIGS. **19-20**, protrusions **1443**, **1445** on arms **1442**, **1444** are sized for engagement within notches, e.g., **1638A**, **1638B** on implant **1600**, on corner surfaces at the sides of an intervertebral implant while head **1415** of inner shaft **1413** is disposable within a complementary cavity within an appropriately sized implant, e.g., implant **1600**. Further, head **1415** may have elastically deformable physical properties to provide for bending of head **1415** when pressed against surfaces so that head **1415** may be advanced through passages smaller than dimensions of the head, such as a cannulation through a locking element, e.g., locking element **1650**. In some examples, implant **1600** may include threads on an interior surface adjacent to locking element **1650**, the threads being complementary to threads on head **1415** of inner shaft **1413**. In some examples, inner shaft **1413** may have both a smaller size than head **1415** and less flexibility while still being translatable through locking element **1650** in implant **1600**.

(87) In another embodiment, an insertion instrument appears as shown in FIGS. **15** and **16**. Unless otherwise noted, like reference numerals refer to like elements of the insertion instrument **1300**, but within the **1500** series of numbers. Insertion instrument **1500** is adapted for use with a variety of implant sizes although does not include openings to provide for fastener insertion while engaged to an implant.

(88) The insertion instrument may be varied in many ways. For example, the distal end region of the outer shaft and/or intermediate shaft may include prongs adapted to engage interior walls of an implant where the implant does not include a locking element disposed therein. This design may be desirable where a locking element to be used is not cannulated.

(89) In yet another aspect, the present disclosure relates to a system including an insertion instrument and an intervertebral implant. In some embodiments, an insertion instrument such as those described herein may be used in conjunction with any implant having a vacant central opening so that longitudinally extending prongs of the insertion instrument may be inserted therein. For instance, with embodiments of the implant having an opening through the locking element. In other embodiments, a system includes an insertion instrument, an intervertebral implant and a drill guide structure, such as drill guide **1350** shown in FIG. **14B**. Drill guide **1350** includes a longitudinally extending engagement feature (not shown) along its length to engage with an insertion instrument and may be held in place axially through a back stop provided by button **1320**, again, as shown in FIG. **14B**. Drill guide **1350** includes guide holes **1352**, **1354** protruding from a main body and angled in opposite directions, the angulation corresponding to an angulation of openings in the implant attached to the insertion instrument.

(90) In yet another aspect, the present invention relates to a method of manufacturing an intervertebral implant. In some embodiments, the implant is formed using an ALM fabrication process such as SLS, SLM or EBM described above, fused deposition modeling (FDM), or another appropriate 3D printing technology known to those skilled in the art. When employing powder-bed based technologies, articles are produced in layer-wise fashion according to a predetermined digital model of such articles by heating, e.g., using a laser or an electron beam, multiple layers of powder, which may be a metallic powder, that are dispensed one layer at a time. The powder is sintered in the case of SLS technology and melted in the case of SLM technology, by the application of laser energy that is directed in raster-scan fashion to portions of the powder layer corresponding to a cross section of the article. After the sintering or melting of the powder on one particular layer, an

additional layer of powder is dispensed, and the process repeated, with sintering or melting taking place between the current layer and the previously laid layers until the article is complete. The powder layers similarly may be heated with EBM technology. Additive manufacturing techniques such as the ALM processes described above may be employed to form the implant, the locking element, or the combination of the implant and the locking element. It should also be appreciated that other devices, instruments, or components therefor that are contemplated by this disclosure may also be formed through an ALM process. In some instances, materials for one layer may be different than the materials for successive layers.

(91) In some embodiments, an implant and accompanying locking element are formed through a single, continuous ALM process. Put another way, the combined structures are formed layer by layer through a single step. This approach may be employed to form any one of implants **100, 200, 300, 400, 800, 900**, for example. The geometry and other properties of the implant and the locking element are programmed into software associated with the ALM system, e.g., computer and machine, and then used to produce both elements together in a single pass. When complete, the locking element is located either on the implant (e.g., implants **100, 800, 900**) or in the implant (e.g., implants **200, 300, 400**).

(92) In some embodiments, an implant is formed through a two-step ALM process where a first part of the implant is formed via ALM, then paused, and then later resumed to completion. This ALM process is focused on the implant itself and does not include formation of the locking element, as described in greater detail below. The two-step approach is particularly well suited for implants with a locking element enclosed within the implant. More particularly, where the locking element is not formed through ALM, this approach facilitates the inclusion of such locking elements in the implant even when the locking element is disposed in an internal cavity of the implant without openings large enough to insert or remove the locking element. The two-step process has particular applicability as an option for implants **200, 300, 400, 500, 600, 700, 800** for example.

(93) Turning to the details of the process, a first portion of an implant is formed through an ALM technique. With reference to implant **200** as illustrative, printing begins from leading surface **204** and continues toward an opposite end of the implant until just shy of trailing wall **203**. These locations on implant **200** are best shown in FIGS. 2A and 2B. To perform the first step, a mold may be used to keep the partially formed implant in a fixed location. Upon reaching a location of the implant structure approximately where head **252** of locking element **250** would be disposed, printing is stopped. Then, locking element **250** is retrieved and disposed in a formed portion of cavity **211** of implant **200**. Locking element **250** may be formed by any desired method, either subtractive or additive. Of course, as noted above, locking element **250** may also be formed together with the implant body in a single continuous process. Locking element **250** is checked to confirm it is positioned within cavity **211** at a sufficient depth, then step two of the process may commence. ALM resumes and the remainder of the implant is printed, including trailing wall **203**. In each of the first and second steps, the same pre-programmed model is used to print the implant, so the final implant, even with the break in printing, still reflects the desired properties and dimensions. Indeed, the final implant is a monolithic structure. This method may be similarly performed for implants **300, 400, 500, 600, 700**. For implant **800**, a first step involves printing implant **800** from surface **804** to a location on the implant aligned with the narrower portion of the locking element. Then, printing is paused and locking element **850** is inserted from opening **813** into its intended position within the implant. To continue, locking element may need to be temporarily held in place so that it does not become displaced. The second step is then performed by printing the remainder of implant **800** up to surface **802** so that locking element **850** is entirely held within implant **800**, as shown in FIG. 8B.

(94) In yet another embodiment, the implant itself may be formed in its entirety through ALM in a continuous single step process without the locking element. In this method, the locking element

may be inserted into the implant after formation of the implant via ALM. In some cases, an implant may simply be inserted into a cavity on a trailing surface of the implant, such as with implants **100**, **900**. In other examples, the cavity for the implant is blocked from the trailing side. In these examples, the implant includes access passages, i.e., slots, so that the locking element may be slid into the cavity within the implant and secured in place with a pin, for example. Examples of the latter configuration include implants **500**, **600**, **700**. With reference to implant **500** as illustrative, implant **500** is first formed through a single step continuous ALM process, layer by layer. Then, locking element **550** is slid into cavity **511** through either side access **526** or side access **527**, as shown in FIG. 5D. Insertion may involve first orienting locking element **550** so its long dimension is perpendicular to a length of the side access opening edge and then advancing it into one of openings **522**, **524** before pushing it into one of the side access openings. Once locking element **550** is centered on central longitudinal axis **582**, pin **570** is used to secure locking element **550** in place. In this method, locking element **550** may be formed through ALM or any other manufacturing process.

(95) Materials used to form the implant, locking element and/or various components described above with an ALM process include, but are not limited to, metals (e.g., metal powder) that may be any one or any combination of titanium and its alloys, stainless steel, magnesium and its alloys, cobalt and its alloys including cobalt chromium alloys, nickel and its alloys, platinum, silver, tantalum niobium, and other super elastic materials such as copper-aluminum alloys. Of the aforementioned examples, titanium is particularly well suited for the implants described herein. Non-metallic materials may also be used and include, but are not limited to, implantable plastics. These may be any one of or a combination of wax, polyethylene (PE) and variations thereof, polyetheretherketone (PEEK), polyetherketone (PEK), acrylonitrile butadiene styrene (ABS), silicone, and cross-linked polymers, bioabsorbable glass, ceramics, and biological active materials such as collagen/cell matrices. Of the aforementioned examples, PEEK is particularly well suited for the implants described herein. Combinations of material types are also contemplated. For example, the implant may be formed of a titanium coated PEEK. To the extent other materials are described elsewhere in the specification, such materials are also contemplated for use in ALM processes.

(96) In yet another aspect, the present disclosure relates to a method of implanting an intervertebral implant using an implant insertion instrument. In some embodiments, insertion instrument **1300** is used to perform the method. For purposes of illustration, the method will be described with reference to implant **100**, the details of which are shown in FIGS. 1A-1G, for placement into an intervertebral space. Initially, insertion instrument **1300** is engaged with implant **100**, as shown generally in FIG. 21. Because locking element **150** is built into central longitudinal opening **111** of implant during manufacture, i.e., prior to surgery, as shown in FIG. 1B, no additional step is required during surgery to engage implant **100** with insertion instrument.

(97) To engage the insertion instrument with implant **100**, arms **1342**, **1342** are advanced into engagement with notches **138A**, **138B** on sides of trailing surface **102**. Engagement of the arms to the sides of implant stabilizes the implant relative to the insertion instrument and prevents relative rotation. At the same time, a distal end of outer shaft **1311** is advanced into a cavity **153** within locking element **150** disposed in implant **100**, a step illustrated in FIG. 17. Then, lever arm **1334** is rotated toward the user to push inner shaft **1313** within outer shaft **1311**, advancing inner shaft **1313** as represented in a comparison of FIGS. 17 and 18. As inner shaft **1313** passes between longitudinally extending prongs **1314A**, **1314B** at a distal end of the outer shaft, exterior surfaces of the prongs press against the inner walls of shaft **166** of the locking element. In turn, pressure is applied from shaft **166** onto walls of central longitudinal opening **111**. Thus, in the engaged position, pressure from prongs **1314A**, **1314B** against locking element **150** holds implant **100** in place relative to insertion instrument **1300**, while arms **1342**, **1344** prevent relative rotation between the implant and the insertion instrument.

(98) At this juncture, the insertion tool is used to advance the implant into a prepared intervertebral space between vertebrae **2**, **4**, as shown in FIG. **22**. Then, fasteners are inserted into respective openings **122**, **124** in implant **100**, as shown in FIG. **23**. Optionally, a drill guide, such as drill guide **1350**, may be slid onto insertion instrument **1300** into a position shown, for example, in FIG. **14B**, prior to the fastener insertion step to provide additional guidance for the fastener insertion. In either variation, the fasteners may be advanced into respective vertebrae adjacent to the implant with the insertion instrument engaged to the implant, as shown in FIG. **23**. This is made possible by the angled open pathways **1322**, **1324** built into distal region **1317** of insertion instrument **1300**, and, by extension, the guide paths **1352**, **1354** built into drill guide **1350**, as applicable. Due to the trajectory of openings **122**, **124** in implant **100**, each fastener may be anchored at a steep angle to provide for anchorage into an outer corner of a respective vertebra when inserted therein.

(99) Once the user is satisfied with the implant position in the intervertebral space and the fastener anchorage, instrument **1300** is removed and locking element **150** is rotated into a blocking position, as shown in FIG. **24**, to prevent back out of fasteners (fasteners not shown in FIG. **24**). With respect to implant **100** specifically, a driver is used to engage drive element **155**, and locking element **150** is rotated ninety degrees to shift the end portions of locking element **150** from a superior-inferior orientation outside of the path of openings **122**, **124** to a sideways oriented direction whereby each end portion blocks a respective opening **122**, **124**. The locking element provides a visual of the partial coverage of the fastener heads in openings **122**, **124**. Also, the locking element provides a tactile and audio feedback as it is rotated into a new locked position. For example, as the locking element is rotated into the blocking position, it makes a snap sound as it locks into place, and the user can also feel, through the driver engaged to the locking element, that the locking element is moved into a new locked position. Some or all of these visual, audio and tactile feedback mechanisms are similarly present in other implants described in the present disclosure. This completes the implantation of implant **100**. The method may be similarly performed with implant **300**.

(100) The method of implantation may be varied in many ways. With continued reference to insertion instrument **1300**, in some examples, a locking element is inserted into the implant and secured with a pin during manufacture prior to use of the implant in surgery. For instance, with implants **500**, **600**, **700**, the locking element may be slid into the implant from a side access, e.g., **526**, **527** shown in FIG. **5D**, that is accessible from a fastener opening in the implant. With implants **600**, **700**, the locking element may be inserted through an inferior access in the implant, e.g., **629**, **729**, shown in FIGS. **6D** and **7D**. Then, a pin may be inserted through the central opening in the locking element to secure the locking element in place. For implant **900**, locking element **950** and pin **970** may be snapped into place during manufacture of the implant. In other examples, such as with implants **200**, **300**, **400**, **800**, the implant may be formed with the locking element together in an additive manufacturing process so there is no separate locking element insertion step.

(101) In another example, locking element **1750** is inserted into implant **1700**. Initially, locking element **1750** is advanced toward trailing end **1702** of implant, as shown in FIGS. **25A-B**. An instrument may be used to direct the locking element to the implant through insertion of arms on the instrument through openings **1753**, **1754**. Alternatively, the locking element may be engaged with another instrument or may be held by hand and placed into the implant. As locking element **1750** contacts trailing surface **1702**, chamfers **1761**, **1762** on locking element **1750** contact implant chamfers **1793**, **1794**. As these surfaces contact one another and the locking element is advanced toward recessed surface **1791**, side portions **1751**, **1752** flex inward to slide over the lip around the recessed surface of the implant, and then locking element **1750** snaps into place within recessed surface **1791**. In particular, side portions **1751**, **1752** snap into grooves **1788**, **1789**, as shown in FIGS. **26A-B**. When engaged in grooves **1788**, **1789**, side portions **1751**, **1752** are in tension, thereby holding locking element **1750** in place within implant **1700**. In this position, loop portions **1756**, **1758** are positioned to block openings **1722**, **1724**. During use of the implant during surgery,

screws may be inserted through openings **1722**, **1724** prior to engagement of locking element **1750** to implant **1700**.

(102) In yet another variation of the method, applicable to implants **200**, **400**, **500**, **600**, **700**, **800**, **900**, the locking element may be modified to include a central longitudinal cavity for gripping by insertion instrument **1300** to insert the instrument into the implant. For implants that include a pin to secure the locking element, the pin may be enlarged and include a cavity therein for insertion of prongs **1314A**, **1314B**. In other examples, when an implant from among implants **200**, **400**, **500**, **600**, **700**, **800**, **900** is used but is not modified for insertion of the prongs of instrument **1300**, another instrument may be used to hold the implant and place it into the intervertebral disc space. For instance, an alternative instrument may be forceps or another insertion device with gripping arms structured for engagement with the fastener openings in the implant. Thus, if forceps are used to engage with implant **200** and deliver it to a target space in the spine, then the forceps may grip fastener openings **222**, **224** to hold the implant. In this manner, a fully assembled implant, here, implant **200** with locking element **250** disposed therein, may be delivered to a target site in a patient even without a custom opening in the locking element for engagement by insertion instrument **1300**. Once implant **200** is in a desired position within the spine, the forceps may be removed and the fasteners implanted through the fastener openings. It should be appreciated that this method may also be used even when use of instrument **1300** is another available option, such as with implant **100**.

(103) In another embodiment, an implant may be implanted using insertion instrument **1400** in the same manner as that described for insertion instrument **1300**. However, with insertion instrument **1400**, widened head **1415** on inner shaft **1413** is advanced through a central opening in a locking element, e.g. locking element **1650** within implant **1600**, and into a widened opening internal to the locking element, as shown FIG. **20**. In some examples, a threaded connection is formed between head **1415** and implant **1600**. In this position, inner shaft **1413** is held in place with respect to the implant. To pass through the opening in locking element, inner shaft may bend in an elastic manner. In some examples, the head may be sized to fit through an opening in locking element **1650**. In some examples, arms **1442**, **1444** may be actuated to lock into notches **1638A-B** through advancement of a sleeve along the arms. In yet another embodiment, an implant may be implanted using insertion instrument **1500**. Once implant is in position within an intervertebral space, instrument **1500** may be removed prior to advancing bone fasteners through the implant and into vertebral bone structures.

(104) In some embodiments, the step of blocking fasteners inserted into an implant may be performed with locking mechanism **1150** instead of a locking element such as locking element **150**, as shown in FIGS. **12A-B**. With implant in place within an intervertebral disc space, a fastener is directed to an opening **1122** in implant **1100**. As the fastener reaches second part **1154** of locking mechanism **1150**, the head of the fastener having a diameter larger than that of second part **1154** causes the second part to be pushed and rotated downward with advancement of the fastener until second part **1154** contacts a wall of opening **1122**, as shown in FIG. **12B**. Engagement feature **1155** engages with second engagement feature **1123** to secure locking mechanism **1150** in position, and the fastener is prevented from back out by first part **1152**, now parallel to trailing surface **1102** of implant **100**.

(105) It should be noted that any of the devices and methods disclosed herein can be used in conjunction with robotic technology. For example, any of the implants described herein can be used with robotic surgical systems to place the implant in a patient. The implant can be manipulated with a robotic system or a robotic arm to rotate or position the implant, and to anchor bone fasteners through the implant during a procedure. Further, any or all of the steps described in the methods for performing an implant placement procedure of the present disclosure may be performed using a robotic system. Similarly, robotics may be used in methods of forming the implant with ALM processes.

(106) Although the disclosure herein has been described with reference to particular embodiments, it is to be understood that these embodiments are merely illustrative of the principles and applications of the present invention. It is therefore to be understood that numerous modifications may be made to the illustrative embodiments and that other arrangements may be devised without departing from the spirit and scope of the present invention as defined by the appended claims.

Claims

1. A method of forming an intervertebral implant using additive layer manufacturing comprising: alternately depositing and heating successive layers of a first source material to form a first portion of a body of the intervertebral implant; alternately depositing and heating successive layers of a second source material subsequent to the formation of the first portion to form a fastener blocker; and alternately depositing and heating successive layers of the first source material subsequent to the formation of the first portion to form a second portion of the body, wherein the depositing and heating steps result in the fastener blocker being captured between the first and second portions such that a head of the fastener blocker is contained within the body, and wherein the fastener blocker is configured to be moveable between a first position where the fastener blocker blocks at least part of an opening in the body and a second position where the fastener blocker does not block any part of the opening.
2. The method of claim 1, wherein upon formation of the second portion, the fastener blocker is irremovably enclosed within the body.
3. The method of claim 1, wherein the additive manufacturing process is continuous such that the body and the fastener blocker are formed together in a single sequence.
4. The method of claim 1, wherein the first source material is different from the second source material.
5. The method of claim 1, wherein at least a portion of the body is formed simultaneously with at least a portion of the fastener blocker.
6. The method of claim 1, further comprising: forming one or more struts on the first portion of the body; and forming the fastener blocker such that the fastener blocker is attached to the one or more struts.
7. The method of claim 1, wherein upon forming the first and second portions of the body, the body includes a superior endplate surface, an inferior endplate surface, and the opening, the opening configured to receive a fastener and extending from a trailing end of the body through one of the superior and inferior endplate surfaces.
8. The method of claim 1, wherein the body is formed to include surfaces such that movement of the fastener blocker between the first position and the second position is accompanied by tactile feedback.
9. A method of forming an implant, the method comprising the steps of: depositing and melting successive layers of a first powder material to form a body of the implant, the body including a fastener hole; and depositing and melting successive layers of a second powder material to form a fastener blocker, the fastener blocker being capable of overlying at least a portion of the fastener hole; wherein portions of the body and the fastener blocker are formed at the same time and the fastener blocker is captured within the body such that an outer wall of the body covers at least part of a head of the fastener blocker.
10. The method of claim 9, wherein the first powder material is different from the second powder material.
11. The method of claim 9, wherein when the body and the fastener blocker are formed, the fastener blocker is disposed in an internal cavity of the body such that the head of the fastener blocker is irremovable from a trailing end surface of the body, the body being an intervertebral implant.

12. The method of claim 9, wherein the depositing and melting successive layers of the second powder material to form the fastener blocker commences after commencement of the depositing and melting successive layers of the first powder material to form the body and is completed before completion of the depositing and melting successive layers of the first powder material to form the body.
13. The method of claim 9, wherein after the body and the fastener blocker are formed, the fastener blocker is rotatable while captured within the body.
14. The method of claim 13, wherein the body is formed to include surfaces such that rotation of the fastener blocker between a first position and a second position is accompanied by tactile feedback.
15. A method of forming an intervertebral implant using additive layer manufacturing comprising: alternately depositing and heating successive layers of a first source material to form a first portion of a body of the intervertebral implant; alternately depositing and heating successive layers of a second source material subsequent to the formation of the first portion to form a fastener blocker; and alternately depositing and heating successive layers of the first source material subsequent to the formation of the first portion to form a second portion of the body, wherein subsequent to the formation of the first and second portions, the body includes a leading surface, a trailing surface opposite the leading surface, a superior surface extending between the leading surface and the trailing surface, an inferior surface opposite the superior surface, a first side surface extending between the superior surface and the inferior surface, and a second side surface opposite the first side surface, and wherein the depositing and heating steps result in the fastener blocker being captured between the first and second portions, the fastener blocker being disposed within the body such that an entirety of the fastener blocker is within a volume defined by the leading surface, the trailing surface, the superior surface, the inferior surface, and the first and second side surfaces, the fastener blocker being spaced apart from the trailing surface.
16. The method of claim 15, wherein the first source material is different from the second source material.
17. The method of claim 15, wherein the fastener blocker is formed to include a head and a shaft, the shaft having a free end surface remote from the head, the free end surface being transverse to a length direction of the shaft and at least partially facing an interior surface of the body.
18. The method of claim 15, wherein the fastener blocker is formed to include a head and a shaft, the head and the shaft being discrete and moveable with respect to one another such that the shaft is removeable from the body and the head is removable from the body when the shaft is removed.
19. The method of claim 15, wherein upon formation of the second portion, the fastener blocker is irremovably enclosed within the body.
20. The method of claim 15, wherein the body is formed to include a fastener-receiving opening extending from the trailing surface to the superior surface or the inferior surface, the fastener-receiving opening being defined in part by a protrusion at the trailing end, the protrusion extending outward from a remainder of the superior surface or the inferior surface.
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